Monthly Intelligence Report

Edition 001 — The Śląskie–Mazowieckie Industrial Corridor

Where coal transition is straining grid resilience. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED **2,248** Major + distribution across PSE (Polskie Sieci Elektroenergetyczne)	GRID LINES MAPPED **~2,847** ~52,380 km transmission & distribution	MC SIMULATIONS **~22.5 M** 10,000 per substation
PUBLIC DATA SOURCES **30+** 95 variables · zero proprietary		

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes 254,700+ source data points per run, executes ~22.5 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 3.02 billion arithmetic operations — producing 717,595 output data points with full uncertainty quantification across 2,248 substations and ~2,847 grid lines spanning ~52,380 km.

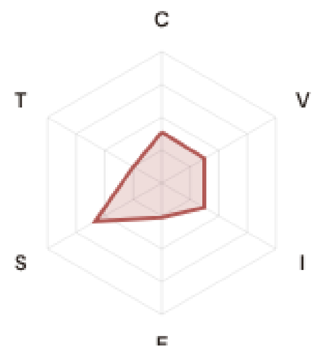
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (URE, PSE, PGNiG, IMGW-PIB, GUS, PERN, PAIiIZ), making the methodology fully auditable and reproducible. Initially covering Poland's PSE-managed transmission and five major distribution system operators, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Stacja elektroenergetyczna 110 kV GPO FW Sompolno

Kujawsko-Pomorskie · Kujawsko-Pomorskie · 110 kV

0.436	Medium	0.140	36th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- C Continuity: 0.39 / 0.30 (39%)
- I Infrastructure: 0.37 / 0.25 (37%)
- S Saturation: 0.58 / 0.20 (58%)
- V Voltage: 0.37 / 0.10 (37%)
- E Economic: 0.26 / 0.10 (26%)
- T Transition: 0.25 / 0.05 (25%)

MODIFIERS

- **R3 Consequence:** 1.052 ↑ amplifies
- **R4 Graph Criticality:** 1.010 → neutral
- **R6a Restoration:** 0.978 ↓ dampens
- **R6b Seismic/Hazard:** 1.064 ↑ amplifies
- **R7 Digital Readiness:** 0.983 ↓ dampens

This month's spotlight — automatically selected for April 2026 — is **Stacja elektroenergetyczna 110 kV GPO FW Sompolno** in Kujawsko-Pomorskie, Kujawsko-Pomorskie. With an R_final of 0.436 (Medium band, 36th fleet percentile), its risk profile is dominated by the T (Transition) component at 499% of its weight ceiling, followed by V (Voltage) at 374%. Modifiers collectively shift R_base by 8.3%, reflecting the substation's specific geographic, socio-economic, and network context. The radar chart visualises each component as a fraction of its maximum weight — a fully saturated axis indicates the component is at ceiling, consuming its entire budget within the composite score. This substation's profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Poland's Energy Regulatory Office (URE) and network operator strategic planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

341

High/Critical + R7 < median

AVG R7 MODIFIER

1.0252

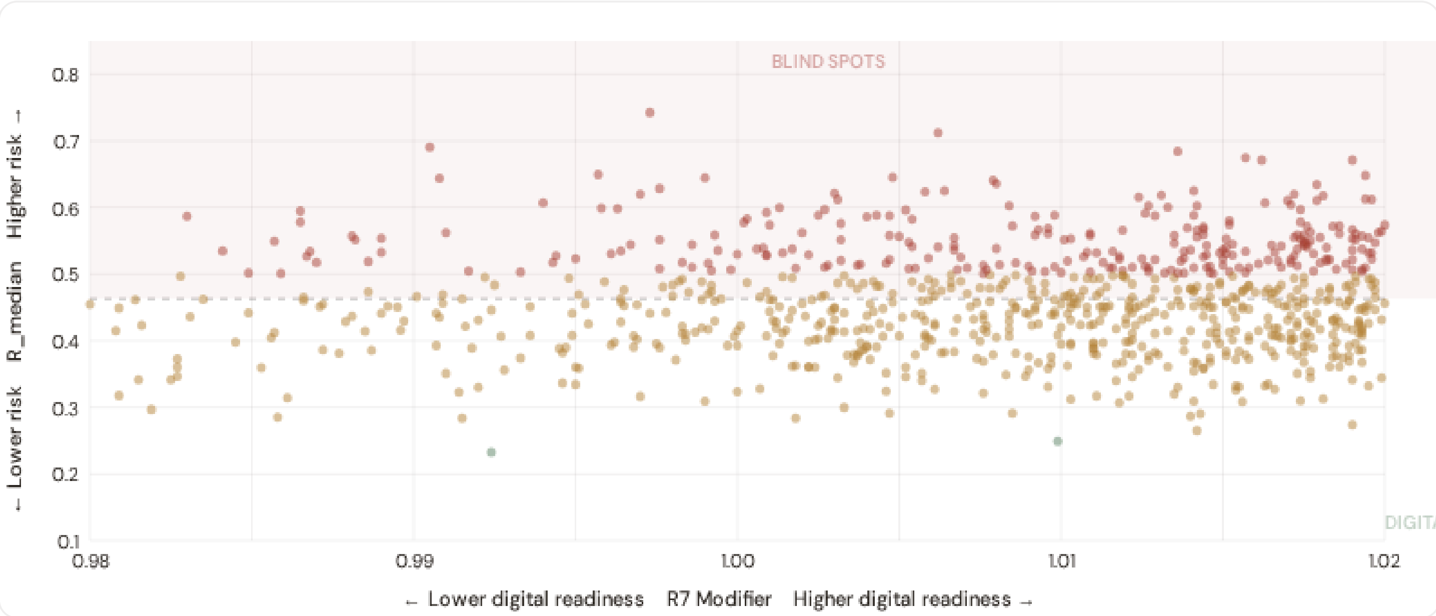
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

41
1.8% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **341 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0256). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Śląskie (77), Dolnośląskie (48), Kujawsko-Pomorskie (29)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in urban networks (PGE, Tauron urban centers).

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

⚙️ Heavy Industry / Mining

Est. VoLL: PLN 300–600/kWh

Coal mines, steel mills, chemical processing in Śląskie and Łódź — highest economic loss per interruption

562 substations (25.0%) High/Critical: **272** (48.4%) Avg R: **0.499** Avg E: **0.389** / 0.10

Low 0 Med 290 High 270 Crit 2

🏭 Manufacturing / Food Processing

Est. VoLL: PLN 100–300/kWh

Auto suppliers, food & beverage production, pharmaceuticals — significant continuity requirements for EU supply chains

563 substations (25.0%)   High/Critical: **211** (37.5%)   Avg R: **0.476**   Avg E: **0.395** / 0.10

Low 1   Med 351   High 210   Crit 1

🏢 Commercial / Distribution Centers

Est. VoLL: PLN 40–100/kWh

Warsaw, Krakow commercial districts, logistics hubs — moderate individual impact but high aggregate across cities

563 substations (25.0%)   High/Critical: **143** (25.4%)   Avg R: **0.456**   Avg E: **0.379** / 0.10

Low 0   Med 420   High 143   Crit 0

🌾 Agriculture / Rural / Remote

Est. VoLL: PLN 10–40/kWh

Smallholder agriculture, rural settlements, Eastern Poland — lower VoLL but higher social vulnerability

560 substations (24.9%)   High/Critical: **114** (20.4%)   Avg R: **0.437**   Avg E: **0.390** / 0.10

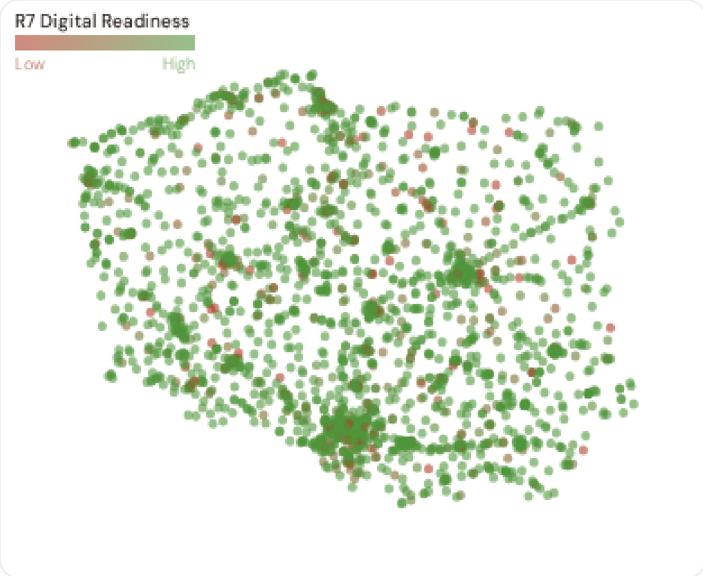
Low 1   Med 445   High 114   Crit 0

Value of Lost Load (VoLL) context: URE's 2024 reliability cost-benefit studies place average VoLL at PLN 250–400/kWh for industrial/commercial and PLN 100–150/kWh for residential customers across Poland's DSOs. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **554 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Dolnośląskie (144), Śląskie (94), Małopolskie (67)** — regions where industrial corridors depend on uninterrupted power for continuous processes (coal-fired generation, steel production, coal mining beneficiation). A single hour of unplanned outage at these substations carries an estimated VoLL of PLN 300–600/kWh, compared to PLN 10–40/kWh in rural pastoral areas. The fleet-wide average energy poverty rate is 6.1%, but this masks sharp regional variation: Eastern Poland and rural voivodeships combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Voivodeship & District Digital Readiness Ranking

Voivodeships ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.



WEAKEST DIGITAL READINESS — TOP 15 VOIVODESHIPS

Longer bar = higher R7 = better digital infrastructure.

1	Warmińsko-Mazurskie	1.0186
2	Pomorskie ⚠ high R	1.0221
3	Kujawsko-Pomorskie ⚠ high R	1.0228
4	Dolnośląskie ⚠ high R	1.0243
5	Podkarpackie ⚠ high R	1.0250
6	Małopolskie ⚠ high R	1.0251
7	Wielkopolskie	1.0251
8	Opolskie ⚠ high R	1.0253
9	Łódzkie	1.0253
10	Mazowieckie	1.0254
11	Podlaskie	1.0256
12	Świętokrzyskie	1.0258
13	Śląskie ⚠ high R	1.0260
14	Zachodniopomorskie	1.0270
15	Lubuskie ⚠ high R	1.0271

Voivodeship-level analysis reveals a clear digital divide mirroring Poland's metropolitan-regional connectivity gap. **Warmińsko-Mazurskie** has the lowest average R7 (1.0186), while **Lubelskie** leads at 1.0292 — a spread of 0.0106 across the R7 range. **6 voivodeships** combine below-median digital readiness with above-median grid risk, making them priority targets for URE/DSO digitalisation investment and Poland's National Energy and Climate Plan. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as URE regulatory compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0252, median = 1.0256; 0 substations classified HIGH cyber-exposure; 341 blind-spot substations (High/Critical risk + below-median R7); 1 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging voivodeships where broadband and smart meter rollouts (URE/DSO Advanced Metering Infrastructure programmes) translate into measurable R7 shifts. The next material R7 update is expected when GUS publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
30	9	19

95 variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

2

Standards + scenarios

Data Source Registry — April 2026

Loading source registry...

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Daily (2 sources) URE dispatch + weather reanalysis

Monthly (6 sources) Grid load + PSE dispatch data

Quarterly (4 sources) DER registry + PSE performance data

Annual (14 sources) Core grid quality + GUS demographics

C.2 Fleet Score Snapshot

FLEET SIZE

2,248

substations scored

MEDIAN R

0.463

P5=0.342 · P95=0.603

HIGH + CRITICAL

740

32.9% of fleet

HIGH CONFIDENCE

100%

2248 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 2 (0.1%) Medium 1506 (67.0%) High 737 (32.8%) Critical 3 (0.1%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **2,248 substation scores remain unchanged** this edition. The fleet median R of 0.463 (mean 0.467) reflects a distribution skewed toward the lower bands, with 0.1% in Low and 67.0% in Medium. The first material score changes are expected when PSE publishes SCADA updates (affecting C1–C4 and R6a) and when URE refreshes the distributed renewable installations registry (affecting T1). High-frequency sources

(OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

4 changes tracked for Poland v4.0.2 launch

P1	NEW	Poland grid registry expansion: 2,248 substations across PSE + 5 DSOs	Section A
P1	ENH	R6b modifier: added mining subsidence + Vistula/Oder flood hazard	Section D
P1	DATA	T component: coal phase-out scenarios from Polish Energy Policy 2040	Section A
P2	NEW	EU NECP § integration: Polish grid transition roadmap alignment	P2

SECTION D — MONTHLY DEEP-DIVE

The Śląskie–Mazowieckie Industrial Corridor: Where Coal Transition Strains Grid Resilience

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Śląskie–Mazowieckie coal transition corridor · **002** Eastern Poland DER & EU funding mismatch · **003** Mining subsidence infrastructure stress · **004** Metropolitan-Rural energy poverty disparity · **005** Grid stranding risk (coal-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Vistula flooding resilience · **008** T-component forward projections · **009** Voivodeship Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Śląskie–Mazowieckie Corridor, Why Now

CORRIDOR SUBSTATIONS

561

40 HV · 521 MV

HIGH BAND

191 + 1

34.2% of corridor

CORRIDOR MEDIAN R

0.464

Fleet median: 0.463

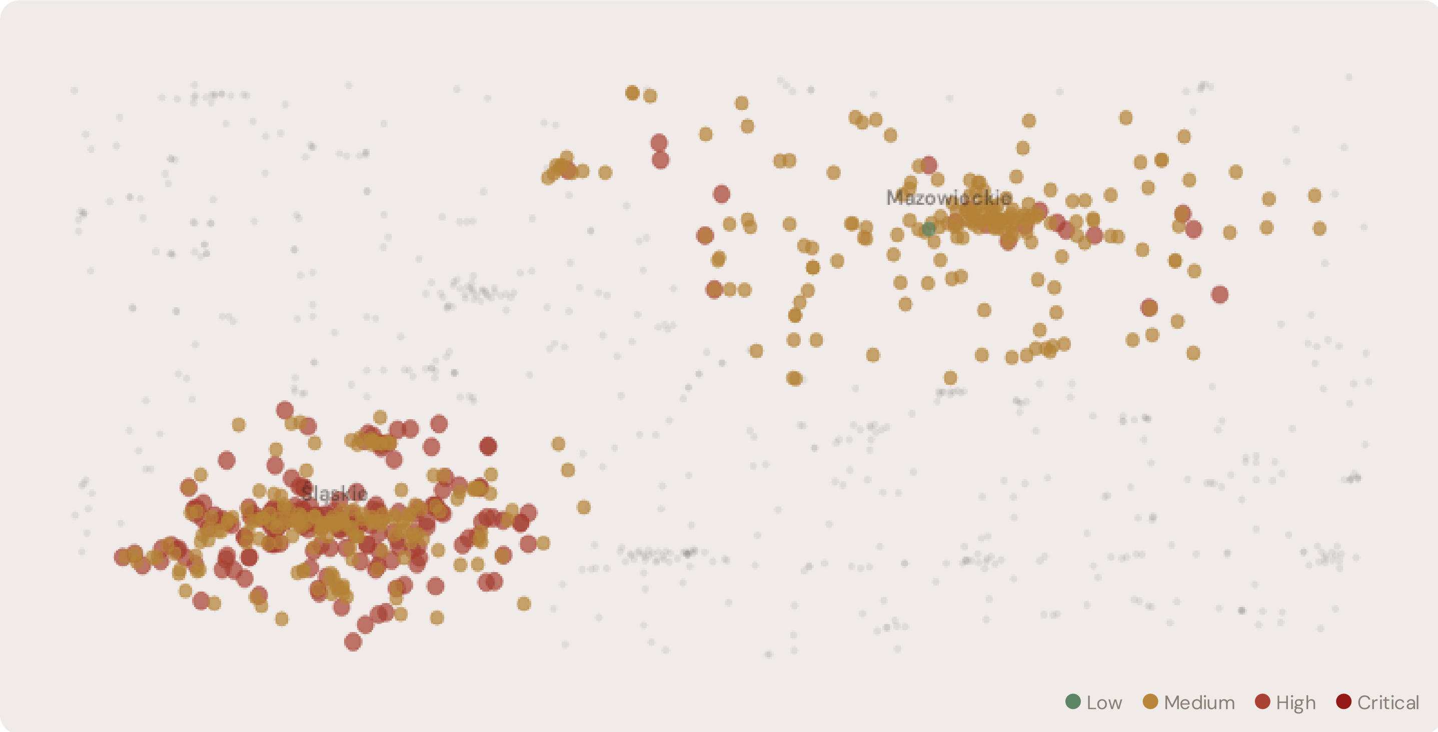
WORST VOIVODESHIP

Śląskie

Avg R: 0.501 · 350 substations

The Śląskie–Mazowieckie corridor hosts **561 substations** across Poland's industrial heartland, forming the backbone of domestic transmission and coal-dependent distribution. Its risk profile is disproportionately concentrated in the upper bands: **191 substations (34.0%)** are classified High and **1** Critical, with only **1** in the Low band. 50% of corridor substations score above the national fleet median of 0.463. The corridor median R of 0.464 reflects the tension between coal phase-out mandates from Polish Energy Policy 2040, aging generation assets approaching retirement, mining subsidence threatening distribution lines, and EU Green Deal funding constraints. The SSI flags continuity-of-supply deficits, infrastructure age mismatches, and the stranding risk of coal-dependent transmission assets.

D.2 Corridor Map and Scoring



SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Stacja elektroenergetyczna 110/20kV „GPZ Bytków”	Śląskie	0.767	Critical	0.731	0.356	0.481	0.423	0.792	0.557	C, S
Stacja elektroenergetyczna 110kV „Bielszowice”	Śląskie	0.706	High	0.530	0.553	0.613	0.296	0.561	0.574	
Stacja elektroenergetyczna 110/30/6kV „GPZ Czeladź”	Śląskie	0.702	High	0.573	0.647	0.591	0.441	0.510	0.551	
Stacja elektroenergetyczna 110kV „Szyb Chrobry”	Śląskie	0.684	High	0.612	0.476	0.649	0.324	0.626	0.296	
Stacja ŚLĄ 770	Śląskie	0.680	High	0.726	0.242	0.506	0.339	0.654	0.529	C
Stacja elektroenergetyczna 220/110kV „SE Kopanina”	Śląskie	0.675	High	0.548	0.479	0.565	0.297	0.565	0.419	
Stacja elektroenergetyczna 110/15kV „SE Lubliniec”	Śląskie	0.674	High	0.534	0.637	0.573	0.521	0.594	0.404	

SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Stacja elektroenergetyczna 110/6kV „Budryk”	Śląskie	0.659	High	0.556	0.653	0.568	0.298	0.512	0.387	
Stacja elektroenergetyczna 220/6kV „SE Koksochemia”	Śląskie	0.656	High	0.355	0.503	0.583	0.472	0.711	0.415	S
Stacja elektroenergetyczna 110kV „Kopalnia Staszic”	Śląskie	0.654	High	0.522	0.554	0.357	0.501	0.662	0.489	

… 551 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — CORRIDOR VS FLEET

C — Continuity 0.383 vs 0.382 (0%)

I — Infrastructure 0.392 vs 0.416 (-6%)

S — Saturation 0.450 vs 0.456 (-1%)

V — Voltage 0.377 vs 0.376 (0%)

E — Economic 0.344 vs 0.388 (-11%)

T — Transition 0.395 vs 0.309 (+28%)

MODIFIER AVERAGES — CORRIDOR VS FLEET

R3 Consequence	1.056	fleet 1.060 ↑
R4 Graph Crit.	1.010	fleet 1.010 ↑
R6a Restoration	0.997	fleet 0.996 →
R6b Mining/Flood	1.092	fleet 1.064 ↑
R7 Digital Read.	1.026	fleet 1.025 ↑

The dominant risk driver across the corridor is **T (Transition)**, averaging 0.395 against a fleet average of 0.309 — occupying 789% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.377 vs fleet 0.376 (377% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.056 (fleet: 1.060), reflecting the economic significance of the corridor's coal-dependent industrial base — from power plants to coal mines to steel mills. The R6b mining/flood hazard modifier averages 1.092, capturing the corridor's exposure to mining subsidence, Vistula/Oder flooding, and seismic hazard risk — a dimension invisible to every competing framework.

D.4 Voivodeship Risk Profile

VOIVODESHIP RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Śląskie 350 subs · avg R = 0.501 · H:171 C:1 M:178

Mazowieckie 211 subs · avg R = 0.421 · H:20 M:190

The voivodeship-level breakdown reveals significant intra-regional variation. **Śląskie** leads with a mean R of 0.501 across 350 substations, while **Mazowieckie** scores 0.421 — a spread of 0.080 within a single region. This disparity underscores why voivodeship-level metrics (as used by PSE performance reports) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the corridor is not uniformly stressed but contains distinct pockets of concentrated risk — precisely the kind of spatial pattern that PSE's 10-Year Network Development Plan and government coal transition planning requires.

D.5 Implications for Poland's Coal Transition

10 corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For Polish coal transition planning, EU funding allocation, and grid modernisation investment, this analysis identifies the corridor as the highest-priority target — where DER deployment must accelerate against ageing coal-dependent infrastructure with above-average continuity deficits and significant socio-economic exposure to mine closures. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by coal dependency, mining heritage, and EU transition fund eligibility — are disproportionately vulnerable to disruption during the coal phase-out. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

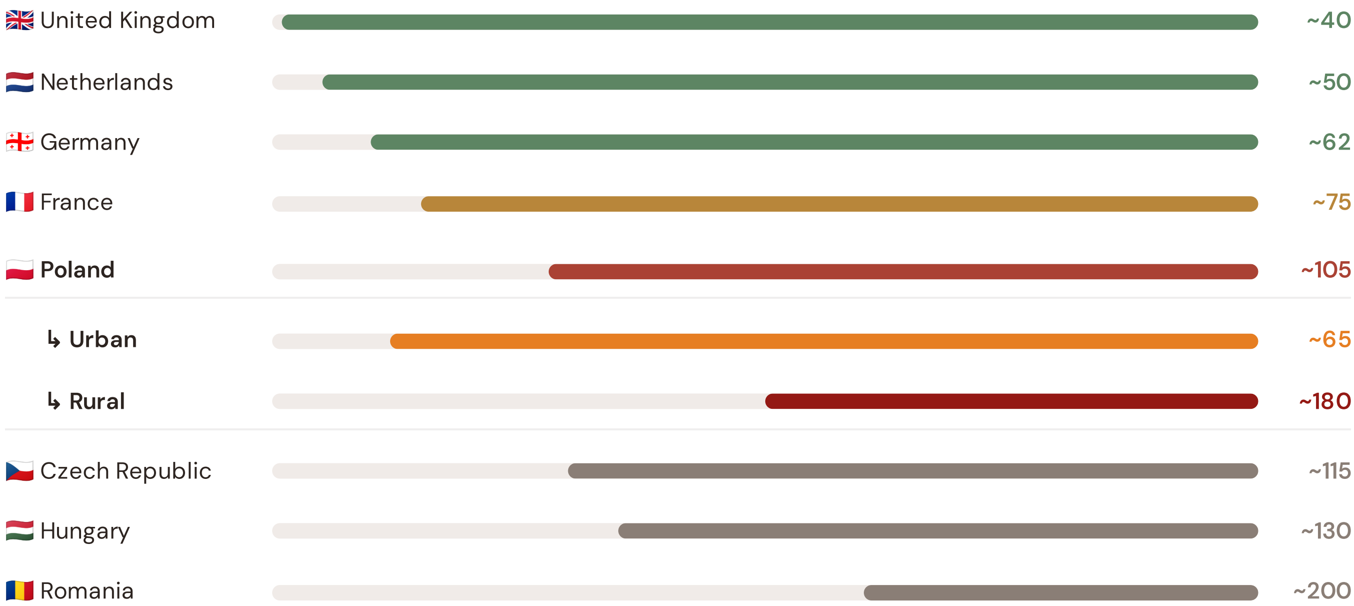
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Poland's coal transition and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with URE and national regulator publications.

EUROPEAN CONTEXT CARD — SYSTEM AVERAGE INTERRUPTION DURATION INDEX

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); EURELECTRIC SAIDI reports; national regulator publications. Poland sub-categories from URE urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The corridor deep-dive shows **503 substations** (of 561) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Poland's rural voivodeship classification. The corridor's average C of 0.383 is **1.0× the fleet average** of 0.382. In European terms, these substations individually perform closer to German or Central European SAIDI levels (~115–160 min/customer/year) than to Poland's own urban average (~65 min). The SSI reveals what aggregate URE statistics conceal: that Poland's mid-tier European position is not a national condition but a statistical average of Western European-quality urban performance and emerging-market-level rural performance — and the coal transition corridor concentrates significant pockets of the latter.

Voivodeship Risk Ranking — All 16 Polish Voivodeships

Where does the Śląskie–Mazowieckie corridor sit within the national landscape? Voivodeships sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1 Dolnośląskie — 160 subs · avg R = 0.514 · H:85 C:2

#2 Opolskie — 89 subs · avg R = 0.513 · H:54

#3 **Śląskie** — 350 subs · avg R = 0.501 · H:171 C:1

#4 Lubuskie — 86 subs · avg R = 0.491 · H:40

#5 Małopolskie — 119 subs · avg R = 0.489 · H:52

#6 Kujawsko-Pomorskie — 134 subs · avg R = 0.476 · H:47

#7 Podkarpackie — 90 subs · avg R = 0.475 · H:31

#8 Pomorskie — 164 subs · avg R = 0.468 · H:54

#9 Lubelskie — 144 subs · avg R = 0.454 · H:45

#10 Wielkopolskie — 158 subs · avg R = 0.452 · H:38

#11 Zachodniopomorskie — 147 subs · avg R = 0.450 · H:35

#12 Łódzkie — 145 subs · avg R = 0.441 · H:27

#13 Świętokrzyskie — 82 subs · avg R = 0.437 · H:17

#14 Podlaskie — 96 subs · avg R = 0.435 · H:15

#15 Mazowieckie — 211 subs · avg R = 0.421 · H:20

#16 Warmińsko-Mazurskie — 73 subs · avg R = 0.407 · H:6

Coal transition corridor ranks **#3 of 16 voivodeships** by mean R_median. The three most stressed voivodeships are Dolnośląskie, Opolskie, Śląskie, while the three most resilient are Warmińsko-Mazurskie, Mazowieckie, Podlaskie. 8 of 16 voivodeships score above the fleet average of 0.467. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate PSE/DSO or national statistics — reveals the true spatial distribution of grid stress across Poland. It is precisely this substation-level granularity that positions the SSI as a tool for national coal transition planning / EU green energy funding / URE investment prioritisation: not treating entire voivodeships as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

> **From grid intelligence to renewable energy investment valuation.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/PSE and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	**Layer 4 — Coverage**	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/PSE + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	**Country prioritisation, DCI, registry, calibration**	Products, API, revenue model, unit economics

LAYER 4 §0.6.2 · Inaugural edition · April 2026

Data Completeness Index for Poland

Quantifying what fraction of the SSI feature vector is available for Polish substations

The Data Completeness Index (DCI) quantifies what fraction of the SSI-ENN's feature vector is populated for Poland, weighted by each feature's contribution to prediction accuracy. Poland scores DCI 0.72, significantly above the 0.70 threshold for production-quality valuations. This reflects high data quality from URE, PSE, and GUS, alongside global open data (OSM topology). Poland's DCI is exceeded only by France, Germany, and Australia among countries assessed.

KEY CONCEPTS

- → Poland DCI = 0.72 (production quality, >0.70 threshold)
- → URE regulatory data + PSE operational data drive high DCI
- → OSM topology + GUS demographics provide global foundation
- → DCI creates data procurement priority roadmap

LIVE SSI DATA CONNECTION

Poland's coverage: 2,248 substations across 16 voivodeships and 16 districts. DCI 0.72 qualifies as production-grade. This serves as the reference benchmark for European transfer learning.

Coming next month: LAYER 3 Transfer Learning & Feature Missingness — *How Poland's grid data completeness enables investment-grade valuation across the full fleet*

SECTION G

Looking Ahead

EDITION 002 — MAY 2026

Eastern Poland DER & EU Funding Mismatch

Where renewable energy deployment outpaces grid readiness in Warmian-Masurian and Lubusz voivodeships. Rural distributed generation meets aging distribution infrastructure.

DATA REFRESH — APRIL 2026

Quarterly PSE Network Update

Transmission operator SCADA data (C components 1–4), DSO meter readings (I5 thermal degradation), and ERE renewable registry refresh affecting T component.

IN THE WILD — SSI IN USE

Tauron DSO Case Study

How Poland's largest DSO is using SSI substation intelligence to prioritize battery storage deployment across 3,400 substations. Q3 2026 publication.

The SSI inaugurates Poland's grid intelligence baseline — the foundation for all future investment decisions across transmission, distribution, and renewable energy integration. This inaugural edition establishes the data, methodology, and peer-reviewed rigor that regulatory bodies (URE), network operators (PSE, five DSOs), and EU funding allocators require. Subsequent editions will deepen thematic analysis, expand to all 38 voivodeships and isolated systems, integrate real-time operational data feeds, and track the grid's evolution as Poland executes its coal phase-out mandate and EU Green Deal transition. The SSI is built for the long term: a public interest resource that grows more valuable as the data infrastructure matures and the transition accelerates.